$= 10\angle 0^\circ$

$5\cos 2t$

$\cancel{10\cos t}$ $10\cos t = \cancel{10\cos t}\ t$

四、电路如图 4.3.7 所示，已知 $u_S(t)=10+10\cos t$ (V)，$i_S(t)=5+5\cos(2t)$ (A)，求 $u(t)$。

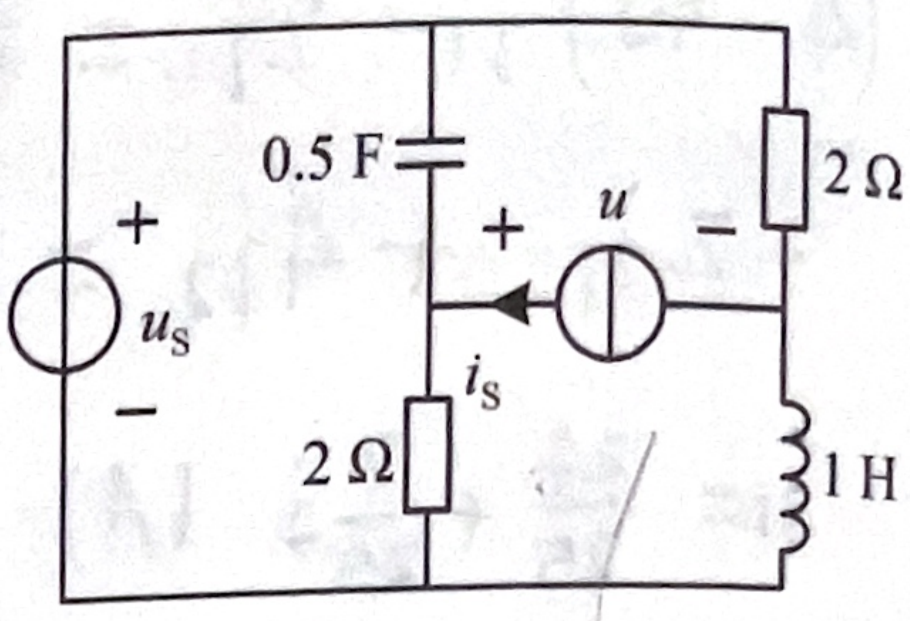
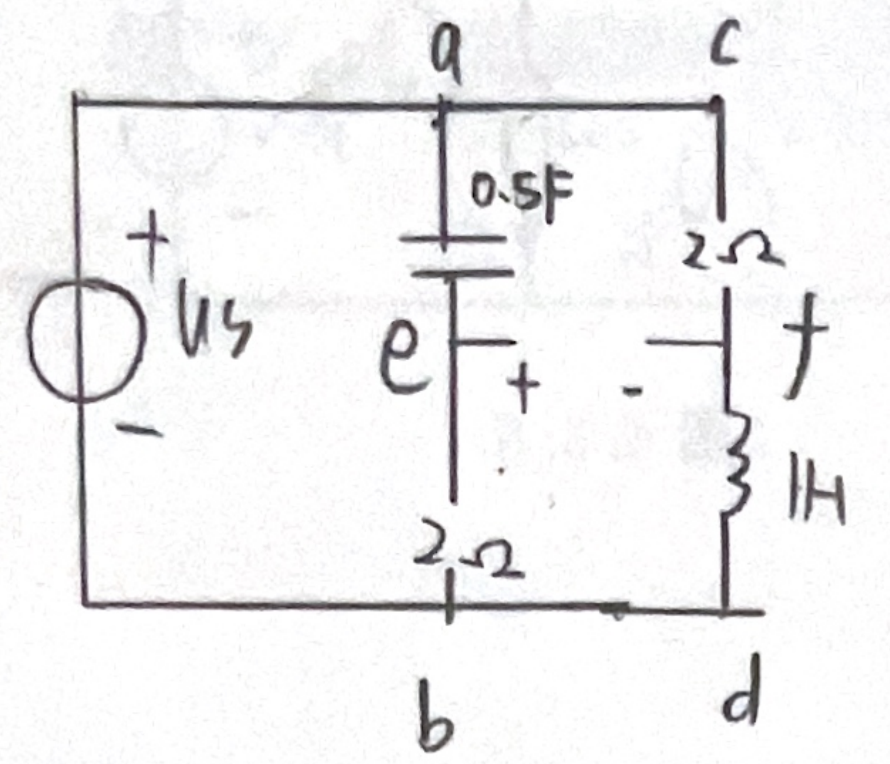


图 4.3.7

解：① $U_{s_1}=10\cos t$



$Z_{ab} = \cancel{\frac{1}{j\omega C}} + R + \frac{1}{j\omega C}$

$= \cancel{2-\frac{1}{2}j} = 2-2j$

$Z_{cd} = R + j\cdot\omega L$

$= 2+j$

$U_{eb} = \frac{2}{2+2-2j}\cdot U_{s_1}$

$= \frac{1}{1-j}\cdot 10\angle 0^\circ$

$= \frac{10}{1-j}$

$= 5(1+j) = 5+5j$

~~② $U_{s_2}=10V$~~

$U_{fd} = \frac{j}{2+j}\cdot U_{s_1}$

$= \frac{j}{2+j}\cdot 10\angle 0^\circ$

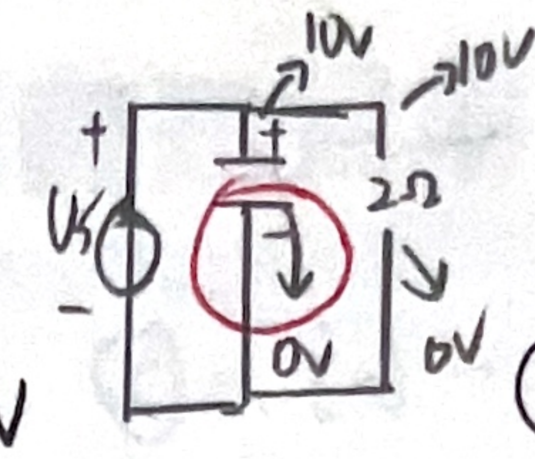
$= 2+4j$

$\therefore U_{ef} = U_{eb} - U_{fd} = (3+j)(V)$

$\therefore U = (3+j)(V)$

$\therefore u(t) = \sqrt{10}\cos(t+18.43^\circ)(V)$

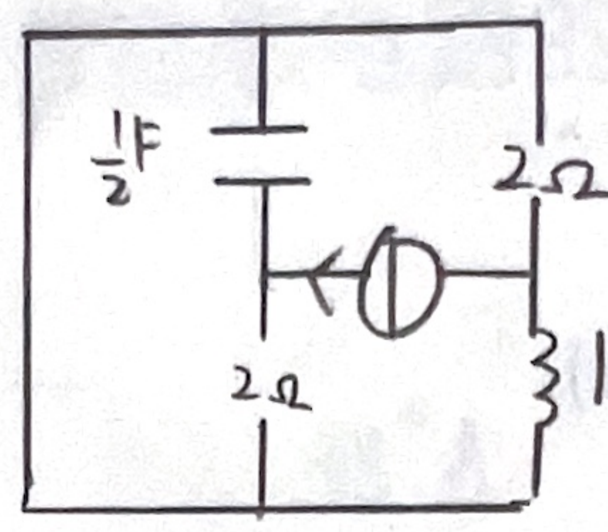
② $U_{s_2}=10$

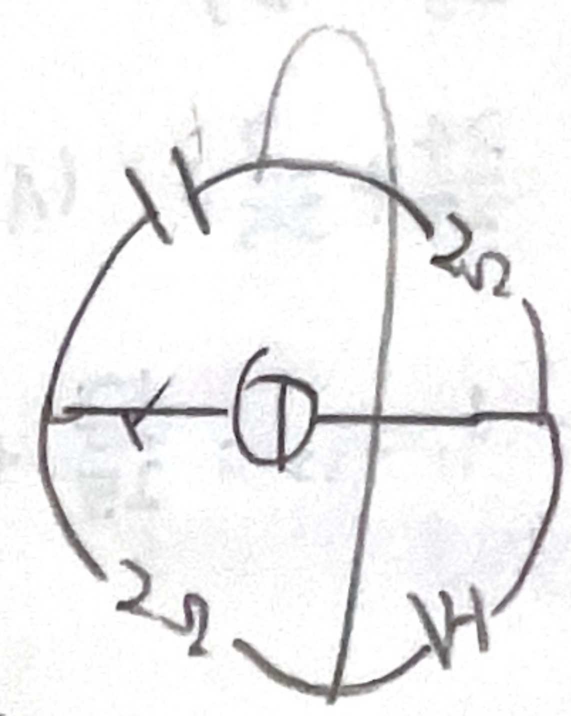


$\therefore U_{ef}=10V$

$\therefore U = \cancel{10V}\ 0V$

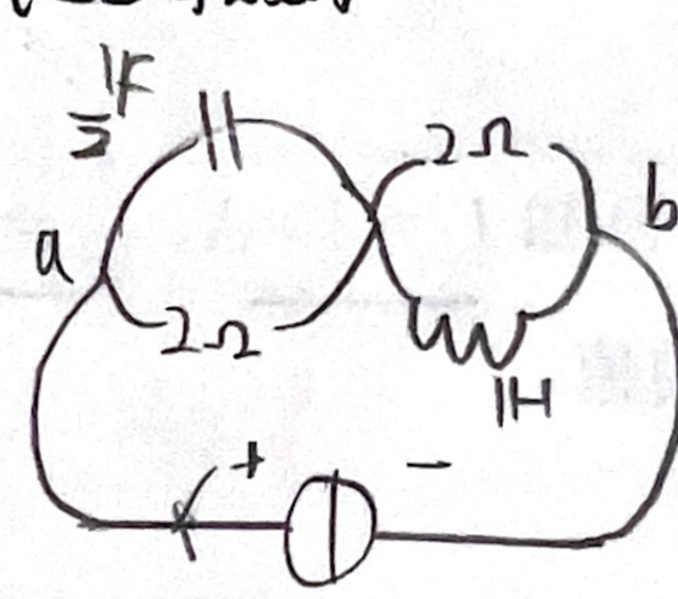
③ $i_{s_1}(t)=5\cos 2t$





~~$i_s = i_{s_1}(t) = 5\cos 2t$~~

电路等效为



$\therefore Z_{ab} = \left[\left(-j\frac{1}{\omega C} \,//\, R\right) + \left(j\omega L \,//\, R\right)\right]$

$= \left[(-j//2) + (2j//2)\right]$

$= \frac{7}{5} + \frac{1}{5}i$

$\therefore U = Z_{ab}\cdot i_{s_1}(t)$

$= \left(\frac{7}{5}+\frac{1}{5}i\right)5$

$= 7+i$

$\therefore u(t) = \sqrt{50}\cos(2t+8.31^\circ)$ (V)

④ $i_{s_2}(t)=5$ A

$U = 10V$

~~$u(t) = 10+\sqrt{10}\cos$~~

~~$u(t) = 10+\sqrt{10}\cos$~~

$\therefore u(t) = \sqrt{10}\cos(t+18.43^\circ) + \sqrt{50}\cos(2t+8.31^\circ) + 10$ V